

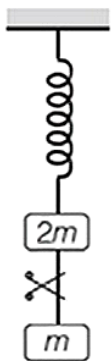
Manzil JEE (2025)

Physics

Laws of Motion

DPP: 6

- Q1** System shown in figure is in equilibrium and at rest. The spring and string are massless, now the string is cut. The acceleration of mass $2m$ and m just after the string is cut will be:-

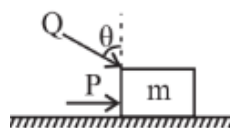


- (A) $g/2$ upwards, g downwards
 (B) g upwards, $g/2$ downwards
 (C) g upwards, $2g$ downwards
 (D) $2g$ upwards, g downwards
- Q2** A 6 kg bomb at rest explodes into three equal pieces P , Q and R . If P flies with speed 30 m/s and Q with speed 40 m/s making an angle 90° with the direction of P . The angle between the direction of motion of P and R is about
 (A) 143°
 (B) 127°
 (C) 120°
 (D) 150°
- Q3** Two identical particles A and B , each of mass m , are interconnected by a spring of stiffness k . If the particle B experiences a force F & the elongation of the spring is x , the relative acceleration between the particles is equal to:

- (A) $\frac{F}{2m}$
 (B) $\frac{F-kx}{m}$
 (C) $\frac{F-2kx}{m}$
 (D) $\frac{kx}{m}$

- Q4** A block of mass m is placed on a smooth wedge of inclination θ . The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block has a magnitude
 (A) $mg \tan \theta$
 (B) $mg \sin \theta$
 (C) $mg \cos \theta$
 (D) $mg \sec \theta$
- Q5** The upper half of an inclined plane of inclination θ is perfectly smooth while the lower half is rough. A body starting from the rest at top comes back to rest at the bottom if the coefficient of friction for the lower half is given by
 (A) $\mu = \sin \theta$
 (B) $\mu = \cot \theta$
 (C) $\mu = 2 \cos \theta$
 (D) $\mu = 2 \tan \theta$

- Q6** A block of mass m lying on a rough horizontal plane is acted upon by a horizontal force P and another force Q inclined at an angle θ to the vertical. The block will remain in equilibrium, if the coefficient of friction between it and the surface is



- (A) $(P + Q \sin \theta)/(mg + Q \cos \theta)$
 (B) $(P \cos \theta + Q)/(mg - Q \sin \theta)$
 (C) $(P + Q \cos \theta)/(mg + Q \sin \theta)$
 (D) $(P \sin \theta - Q)/(mg - Q \cos \theta)$

Q7 Two identical carts constrained to move on a straight line, on which sit two twins of same mass, are moving with equal velocity. At some time snow begins to drop uniformly. Ram, sitting on one the carts, picks the snow from cart and throws off the falling snow sideways and in the second cart Shyam is asleep.

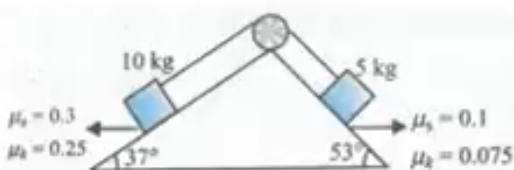
- (A) Cart carrying Ram will have more speed finally than that carrying Shyam.
 (B) Cart carrying Ram will have less speed finally than that carrying Shyam.
 (C) Cart carrying Ram will have same speed finally than that carrying Shyam.
 (D) depends on the amount of snow thrown.

Q8 A bullet of mass 20 g has an initial speed of 1 ms^{-1} , just before it starts penetrating a mud wall of thickness 20 cm. If the wall offers a mean resistance of $2.5 \times 10^{-2} \text{ N}$, the speed of the bullet after emerging from the other side of the wall is close to

- (A) 0.3 ms^{-1} (B) 0.4 ms^{-1}
 (C) 0.1 ms^{-1} (D) 0.7 ms^{-1}

Q9 A system of two blocks and a light string are kept on two inclined faces (rough) as shown in figure. All the required data are mentioned in the diagram. Pulley is light and frictionless. The 10 kg block is initially given a velocity of 2 m/s down the plane (Take

$g = 10 \text{ ms}^{-2}$, $\sin 37^\circ = 3/5$)



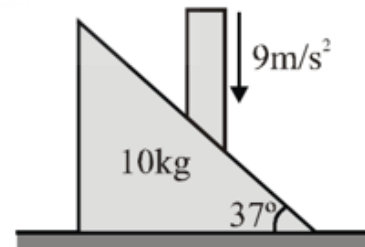
In the above question, the motion of the system would be best described by

- (A) The system first decelerates, comes to a stop, and then continues to move in the opposite direction.
 (B) The system will continuously move with constant speed.
 (C) The system first decelerates and then comes to a stop.
 (D) The system accelerates and its speed increases with time.

Q10 A force of 750 N is applied to a block of mass 102 kg to prevent it from sliding on a plane with an inclination angle 30° with the horizontal. If the coefficients of static friction and kinetic friction between the block and the plane are 0.4 and 0.3 respectively, then the frictional force acting on the block is

- (A) 750 N
 (B) 500 N
 (C) 345 N
 (D) 250 N

Q11 In the system shown in figure, all surfaces are smooth. Rod is moved by external agent with constant acceleration 9 m/s^2 vertically downwards. Force exerted on the rod by the wedge will be :

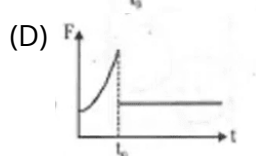
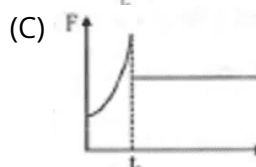
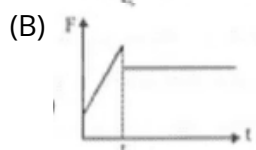
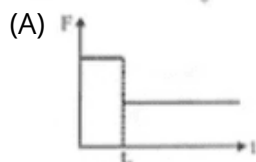
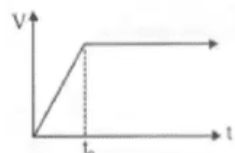


- (A) 120 N (B) 200 N
 (C) 160 N (D) 180 N

Q12 The following graph represents the speed of a car as a function of time. We know that as the car speeds up there is a friction force with air that can be approximately considered to be proportional to the speed of the car. Which of



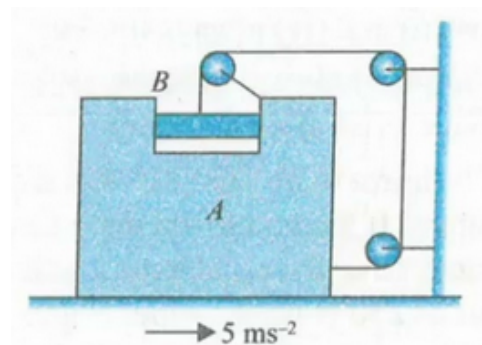
the following graphs can be the force of the engine as a function of time?



Q13 A bead of mass m is free to slide on a fixed horizontal circular wire of radius R . At time $t = 0$, it is given a velocity v_0 along the tangent to the circle. If the coefficient of kinetic friction between the bead and the wire is μ_k . Then magnitude of tangential acceleration at $t = 0$ will be

- (A) $\mu_k g$
 (B) $\mu_k \frac{v^2}{R}$
 (C) $\frac{\mu_k}{R} \sqrt{v^2 + g^2 R^2}$
 (D) $\mu_k \left[g + \frac{v^2}{R} \right]$

Q14 If block A is moving with an acceleration of 5 ms^{-2} , the acceleration of B w.r.t ground is



- (A) 5 ms^{-2}
 (B) $5\sqrt{2} \text{ ms}^{-2}$
 (C) $5\sqrt{5} \text{ ms}^{-2}$
 (D) 10 ms^{-2}

Q15 A block of mass m is placed on a surface with a vertical cross-section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is:

- (A) $\frac{2}{3} m$
 (B) $\frac{1}{3} m$
 (C) $\frac{1}{2} m$
 (D) $\frac{1}{6} m$

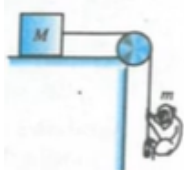
Q16 A smooth light rod not reaching the rough floor is inserted between two identical blocks. A horizontal force F is applied to upper end of rod. The contact force between block A and rod is F_1 and contact force between block B and rod is F_2 . Then,



- (A) $F_1 = F_2$
 (B) $F_1 > F_2$
 (C) $F_1 < F_2$
 (D) $F_1 = 0$ and $F_2 > 0$



- Q17** A monkey of mass m clings to a rope slung over a fixed pulley. The opposite end of the rope is tied to a weight of mass M lying on a horizontal table. The coefficient of friction between the weight and the table is μ . Find the acceleration of weight and the tension of the rope for two cases. The monkey moves downwards with respect to the rope with an acceleration b .

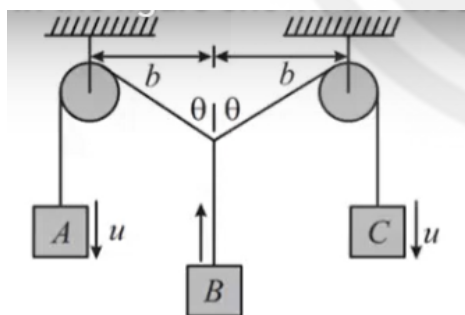


The tension of rope is

- (A) $\frac{Mm(\mu g + g + b)}{M + m}$
 (B) $\frac{Mm(\mu g - g + b)}{M + m}$
 (C) $\frac{Mm(\mu g + g - b)}{M + m}$
 (D) $\frac{Mm(\mu g - g - b)}{M + m}$

- Q18** In the figure shown the blocks A and C are pulled down with constant velocities u .

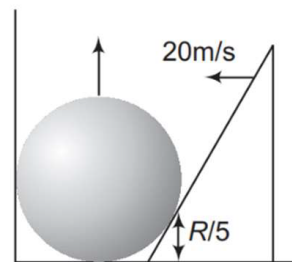
Acceleration of block B is:



- (A) $\frac{u^2}{b} \tan^2 \theta \sec \theta$
 (B) $\frac{u^2}{b} \tan^3 \theta$
 (C) $\frac{u^2}{b} \sec^2 \theta \tan \theta$
 (D) Zero

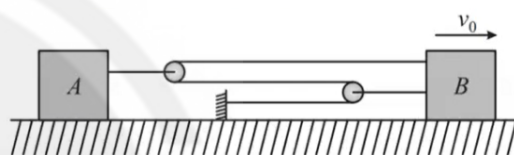
- Q19** A sphere of radius R is in contact with a wedge. The point of contact is from the ground as

shown in the figure. Wedge is moving with velocity 20 ms^{-1} towards left then the velocity of the sphere at this instant will be



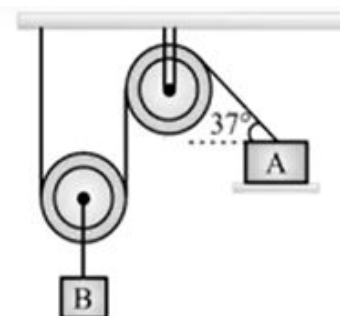
- (A) 20 ms^{-1} (B) 15 ms^{-1}
 (C) 16 ms^{-1} (D) 12 ms^{-1}

- Q20** Block B moves to the right with a constant velocity v_0 . The velocity of block A relative to B is



- (A) $\frac{v_0}{2}$, towards left
 (B) $\frac{v_0}{2}$, towards right
 (C) $\frac{3v_0}{2}$, towards left
 (D) $\frac{3v_0}{2}$, towards right

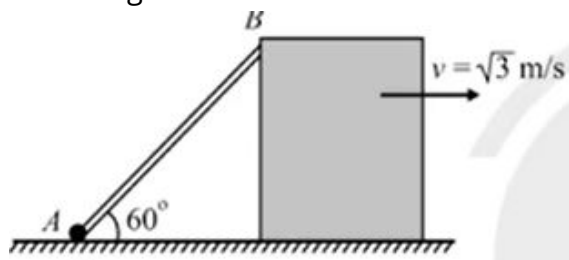
- Q21** In the figure shown block B moves down with a velocity 10 m/s . The velocity of A in the position shown is



- (A) 12.5 m/s
 (B) 25 m/s
 (C) 6.25 m/s
 (D) None of these

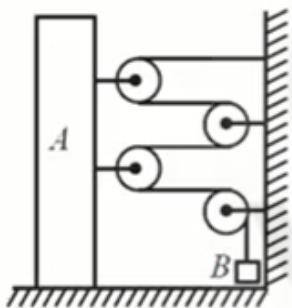


- Q22** A rod AB is shown in figure. End A of the rod is fixed on the ground. Block is moving with velocity $\sqrt{3}$ m/s towards right. The velocity of end B of rod when rod makes an angle of 60° with the ground is



- (A) $\sqrt{3}$ m/s
(B) 2 m/s
(C) $2\sqrt{3}$ m/s
(D) 3 m/s

- Q23** Block A is moving away from the wall at a speed v and acceleration a



- (A) Velocity of B is v with respect to A
(B) Acceleration of B is a with respect to A
(C) Acceleration of B is $4a$ with respect to A
(D) Acceleration of B is $\sqrt{17}a$ with respect to A

- Q24** Velocity of three particles A, B and C varies with

time t as $\vec{v}_A = (2\hat{i} + 6\hat{j})$ m/s

$\vec{v}_B = (3\hat{i} + 4\hat{j})$ m/s and

$\vec{v}_C = (6\hat{i} - 4\hat{j})$ m/s. Regarding the pseudo force match the following table:

List-I

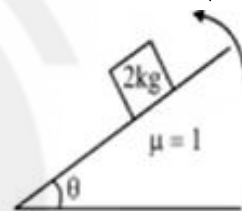
- P. On A as observed by B
Q. On B as observed by C
R. On A as observed by C
S. On C observed by A

List-II

1. Along positive x-direction
2. Along negative x-direction
3. Along positive y-direction
4. Along negative y-direction
5. zero

- (A) P-5, Q-3, R-3, S-2
(B) P-3, Q-1, R-2, S-5
(C) P-4, Q-5, R-2, S-1
(D) P-4, Q-1, R-5, S-3

- Q25** Angle θ is gradually increased as shown in figure. For the given situation match the following two columns, ($g = 10 \text{ ms}^{-2}$)



	Columns-I		Columns-II
A	Force of friction when $\theta = 0^\circ$	P	10 N
B	Force of friction when $\theta = 90^\circ$	Q	$10\sqrt{3}$ N
C	Force of friction when $\theta = 30^\circ$	R	$\frac{10}{\sqrt{3}}$ N
D	Force of friction when $\theta = 60^\circ$	S	None of the above

- (A) A - S, B - S, C - P, D - P
(B) A - S, B - P, C - S, D - P
(C) A - P, B - P, C - R, D - Q
(D) A - S, B - Q, C - R, D - P



Answer Key

Q1 A
Q2 B
Q3 B
Q4 D
Q5 D
Q6 A
Q7 D
Q8 D
Q9 C
Q10 D
Q11 B
Q12 B
Q13 C

Q14 C
Q15 D
Q16 C
Q17 C
Q18 B
Q19 B
Q20 B
Q21 B
Q22 B
Q23 D
Q24 A
Q25 A



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